# Markov Decision Processes

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# 1 : MDP

MDP **(fully observable)** . “ ” Markov , **(reward)** **(action)** .

$$\underbrace{\langle S, P \rangle}_{\text{Markov Process}} \xrightarrow{+R,\,\gamma} \underbrace{\langle S, P, R, \gamma \rangle}_{\text{MRP}} \xrightarrow{+A} \underbrace{\langle S, A, P, R, \gamma \rangle}_{\text{MDP}}$$

> **!**
>
> **MDP** $\pi$ **MRP** , **MRP** **Markov Process** . (evaluation) “ MRP ” .

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# 2 Markov Process

## 2.1 Markov

$$\mathbb{P}[S_{t+1} \mid S_t] = \mathbb{P}[S_{t+1} \mid S_1, \dots, S_t]$$

$S_t$ **(sufficient statistic)** . history .

## 2.2

$$P_{ss'} = \mathbb{P}[S_{t+1} = s' \mid S_t = s], \qquad \sum_{s'} P_{ss'} = 1 \ ( \quad = 1)$$

**Markov Process (Chain)** $\langle S, P \rangle$ — $S$ $P$ . .

## 2.3 Student Markov Chain

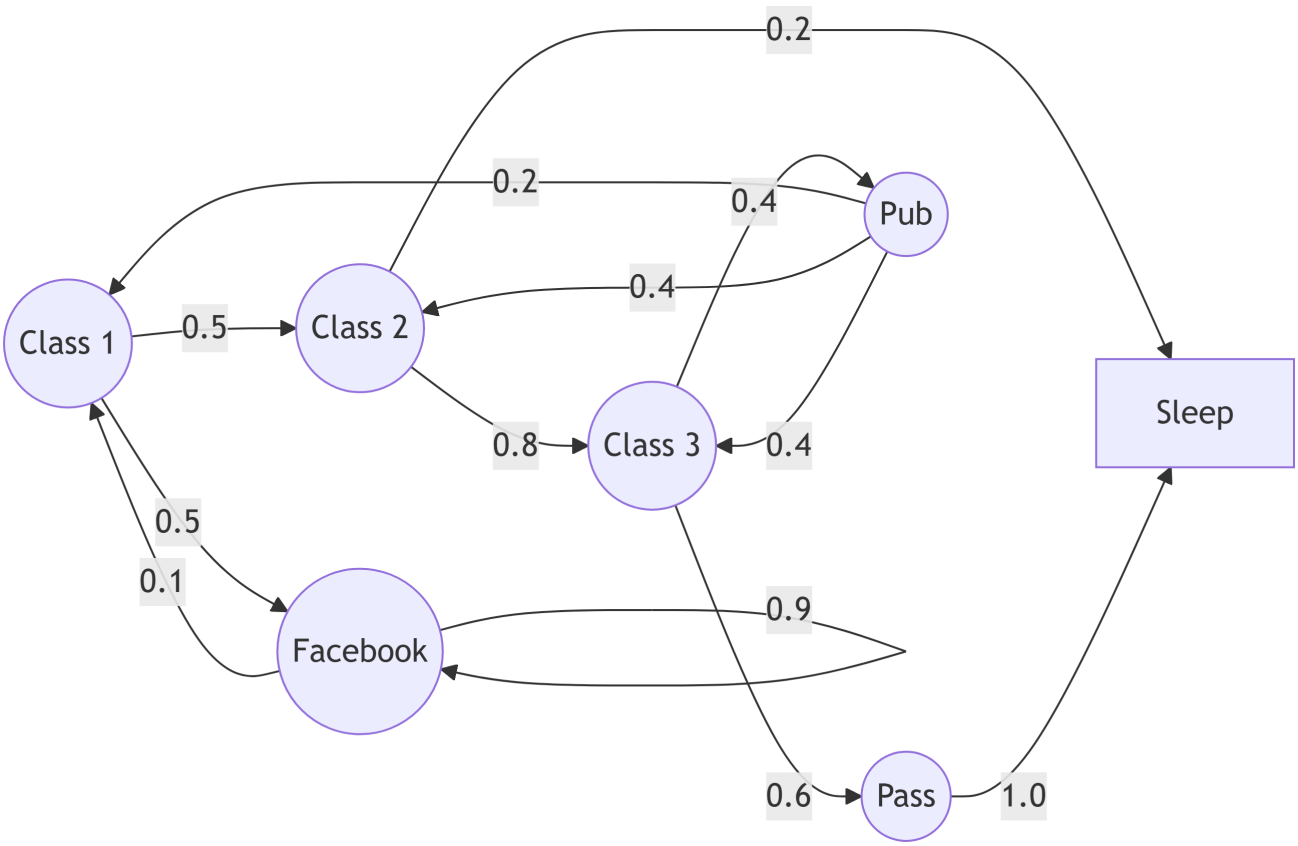
(sample episode) : `C1 → C2 → C3 → Pass → Sleep`, `C1 → FB → FB → C1 → C2 → Sleep` .

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# 3 Markov Reward Process (MRP)

MRP $= \langle S, P, R, \gamma \rangle$ — Markov chain **(value)** .

- : $R_s = \mathbb{E}[R_{t+1} \mid S_t = s]$
- : $\gamma \in [0, 1]$




1: Student Markov Chain

### 3.1 Return ( )

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

> **i**
>
> $s$ $R_{t+1}$ . $G_t$ , $v(s)$ .

### 3.2 — $\gamma$

- ( )
- (cyclic) Markov process **return**
-
-
- /
- (terminate) $\gamma = 1$ (undiscounted)

### 3.3 Bellman Equation

$$v(s) = \mathbb{E}[G_t \mid S_t = s]$$

Return + .

$$\begin{aligned} v(s) &= \mathbb{E}[R_{t+1} + \gamma G_{t+1} \mid S_t = s] \\ &= \mathbb{E}[R_{t+1} + \gamma\, v(S_{t+1}) \mid S_t = s] \\ &= R_s + \gamma \sum_{s'} P_{ss'}\, v(s') \end{aligned}$$

$\mathbb{E}[G_{t+1} \mid S_{t+1} = s'] = v(s')$ tower property .

### 3.4

$$v = R + \gamma P v \implies (I - \gamma P)\, v = R \implies v = (I - \gamma P)^{-1} R$$

- , $O(n^3)$ $(n = |S|)$.
- $\gamma < 1$ $\rho(\gamma P) < 1$ $(I - \gamma P)$ . $n$ DP / Monte-Carlo / TD .

### 3.5 $\gamma$

Sleep (terminal, $v = 0$) $\{C1, C2, C3, \text{Pass}, \text{Pub}, FB\}$ $v_T = (I - \gamma Q)^{-1} R_T$ .
$Q$ , $\gamma = 1$ $\rho(Q) < 1$ .

```python
import numpy as np
import matplotlib.pyplot as plt

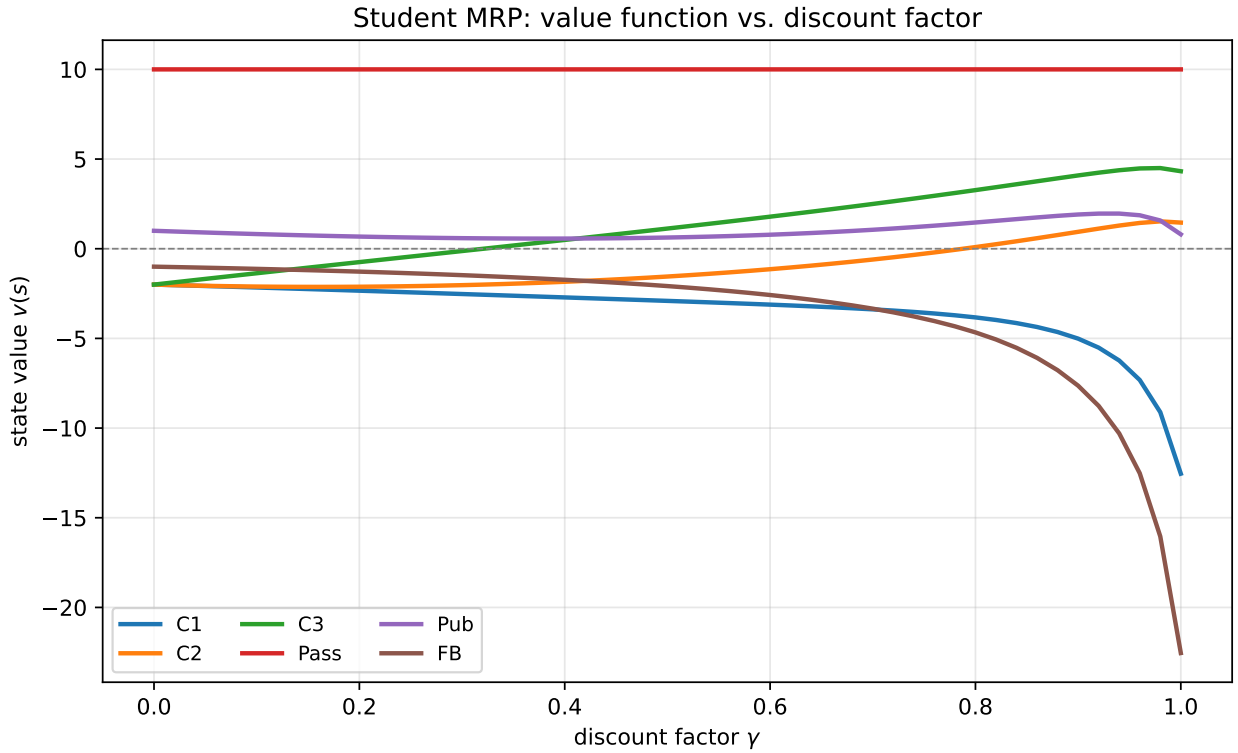
states = ["C1", "C2", "C3", "Pass", "Pub", "FB"]
idx = {s: i for i, s in enumerate(states)}

Q = np.zeros((6, 6))
Q[idx["C1"], idx["FB"]] = 0.5;  Q[idx["C1"], idx["C2"]] = 0.5
Q[idx["C2"], idx["C3"]] = 0.8                      # C2 -> Sleep 0.2 (terminal)
Q[idx["C3"], idx["Pub"]] = 0.4; Q[idx["C3"], idx["Pass"]] = 0.6
# Pass -> Sleep 1.0 (terminal):    0
Q[idx["Pub"], idx["C1"]] = 0.2
Q[idx["Pub"], idx["C2"]] = 0.4
Q[idx["Pub"], idx["C3"]] = 0.4
Q[idx["FB"], idx["FB"]] = 0.9;  Q[idx["FB"], idx["C1"]] = 0.1

R = np.array([-2, -2, -2, 10, 1, -1], dtype=float)

gammas = np.linspace(0.0, 1.0, 51)
V = np.array([np.linalg.solve(np.eye(6) - g * Q, R) for g in gammas])

fig, ax = plt.subplots(figsize=(8, 5))
for j, s in enumerate(states):
    ax.plot(gammas, V[:, j], marker="", linewidth=2, label=s)
ax.axhline(0, color="gray", linewidth=0.8, linestyle="--")
ax.set_xlabel(r"discount factor $\gamma$")
ax.set_ylabel(r"state value $v(s)$")
ax.set_title("Student MRP: value function vs. discount factor")
ax.legend(ncol=3, fontsize=9)
ax.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```




2: gamma v(s)

$\gamma = 0$ $v(s) = R_s$ ( ) . $\gamma$ , $R = +10$ Pass $C3, C2$ . 0.9
FB $\gamma$ $R = -1$ ($\gamma = 1$ $v(FB) \approx -22.5$).

| $\gamma$ | C1 | C2 | C3 | Pass | Pub | FB |
|---|---|---|---|---|---|---|
| 0.0 | −2.00 | −2.00 | −2.00 | +10.00 | +1.00 | −1.00 |
| 0.5 | −2.91 | −1.55 | +1.12 | +10.00 | +0.62 | −2.08 |
| 0.9 | −5.01 | +0.94 | +4.09 | +10.00 | +1.91 | −7.64 |
| 1.0 | −12.54 | +1.46 | +4.32 | +10.00 | +0.80 | −22.54 |

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## 4 Markov Decision Process (MDP)

MDP $= \langle S, A, P, R, \gamma \rangle$ — MRP **(decision)** .

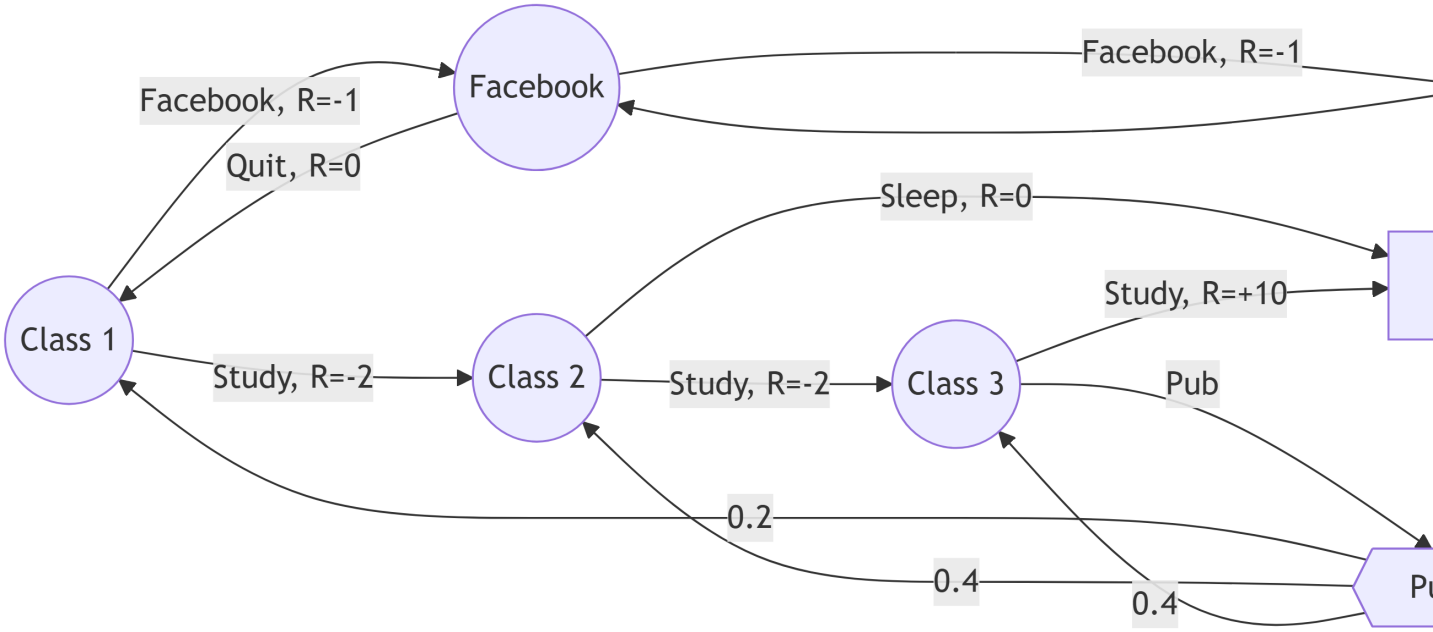
$$P^a_{ss'} = \mathbb{P}[S_{t+1} = s' \mid S_t = s,\ A_t = a], \qquad R^a_s = \mathbb{E}[R_{t+1} \mid S_t = s,\ A_t = a]$$

## 4.1 (policy)

$$\pi(a \mid s) = \mathbb{P}[A_t = a \mid S_t = s]$$

(stationary), . history Markov .

## 4.2 Student MDP




3: Student MDP ( = action / reward)

MRP “ ” , MDP “ ” . Pub (0.2/0.4/0.4) , .

## 4.3 → MRP

$\pi$ $S_1, S_2, \ldots$ Markov process $\langle S, P^\pi \rangle$, - MRP $\langle S, P^\pi, R^\pi, \gamma \rangle$ .

$$P^\pi_{ss'} = \sum_a \pi(a \mid s)\, P^a_{ss'}, \qquad R^\pi_s = \sum_a \pi(a \mid s)\, R^a_s$$

## 4.4

$$v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s], \qquad q_\pi(s,a) = \mathbb{E}_\pi[G_t \mid S_t = s,\, A_t = a]$$

$v_\pi$ , $q_\pi$ ( , ) . .

### 4.5 Bellman Expectation Equation

$v$ $q$ .

$$
\begin{aligned}
v_\pi(s) &= \sum_a \pi(a \mid s)\, q_\pi(s,a) \\
q_\pi(s,a) &= R_s^a + \gamma \sum_{s'} P_{ss'}^a\, v_\pi(s') \\
v_\pi(s) &= \sum_a \pi(a \mid s) \Big( R_s^a + \gamma \sum_{s'} P_{ss'}^a\, v_\pi(s') \Big) \\
q_\pi(s,a) &= R_s^a + \gamma \sum_{s'} P_{ss'}^a \sum_{a'} \pi(a' \mid s')\, q_\pi(s',a')
\end{aligned}
$$

:

$$
v_\pi = R^\pi + \gamma P^\pi v_\pi \quad \Longrightarrow \quad v_\pi = (I - \gamma P^\pi)^{-1} R^\pi
$$

MRP .

---

## 5 (Optimality)

### 5.1

$$
v_*(s) = \max_\pi v_\pi(s), \qquad q_*(s,a) = \max_\pi q_\pi(s,a)
$$

MDP " " $v_*$ ( $q_*$) .

### 5.2

MDP .

1. $\pi_*$ .
2. $v_{\pi_*}(s) = v_*(s)$ .
3. $q_{\pi_*}(s,a) = q_*(s,a)$ .

### 5.3

$q_*$ greedy (deterministic) .

$$
\pi_*(a \mid s) = \begin{cases} 1 & \text{if } a = \arg\max_{a \in A} q_*(s,a) \\ 0 & \text{otherwise} \end{cases}
$$

!

. $q_*$ MDP .

### 5.4 Bellman Optimality Equation

$$v_*(s) = \max_a q_*(s,a)$$
$$q_*(s,a) = R_s^a + \gamma \sum_{s'} P_{ss'}^a \, v_*(s')$$
$$v_*(s) = \max_a \left( R_s^a + \gamma \sum_{s'} P_{ss'}^a \, v_*(s') \right)$$
$$q_*(s,a) = R_s^a + \gamma \sum_{s'} P_{ss'}^a \max_{a'} q_*(s',a')$$

### 5.5 Expectation vs. Optimality

| | | |
|---|---|---|
| Bellman **Expectation** | $\sum_a \pi(a \mid s)\,(\cdot)$ ( ) | $(I - \gamma P^\pi)^{-1} R^\pi$ |
| Bellman **Optimality** | $\max_a(\cdot)$ | $\rightarrow$ |

max , **Value Iteration, Policy Iteration, Q-learning, Sarsa** .

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## 6 Extensions to MDPs

### 6.1 · MDP

- (countably infinite) / :
- / : -2 (LQR)
- : $\rightarrow$ **HJB(Hamilton-Jacobi-Bellman)** , Bellman time-step $\rightarrow 0$

### 6.2 POMDP ( )

$\langle S, A, O, P, R, Z, \gamma \rangle$. MDP = HMM.

- **Belief state** $b(h)$: history $h$
- POMDP ( ) **history tree** ( ) **belief state tree**

### 6.3 Undiscounted / MDP

- **Ergodic** Markov process = recurrent( ) + aperiodic( )
- $\pi$ chain ergodic MDP ergodic
- ergodic MDP $\rho^\pi$
- **average-reward Bellman equation** ( differential value function )

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## 7



1. $v = (I - \gamma P)^{-1} R$ — Bellman . (MRP/leaky bucket · )
2. **Expectation vs. Optimality** **4** — $v \leftrightarrow q$ . max vs. $\sum_a \pi$ / .
3. $\gamma$ **6** .
4. $+ q_*$ greedy .
5. $\rightarrow$ **MRP** $(P^\pi, R^\pi)$ — .

leaky bucket · contention "Markov chain " , .